# George Chen

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## Skills

**Software:** C, C++, Python, VHDL, Verilog, RISC-V, MATLAB, Java, C#, JavaScript, Linux, Git.
**Tools:** Altium Designer, LTspice, KiCad, STM32CubeIDE, Quartus Prime, COMSOL, AutoCAD, Revit.
**Hardware/Lab:** STM32, PCBA Design, Oscilloscope, DMM, Soldering, Function Generator, Logic Analyzer.

## Work Experience

**Electrical Designer - Plan Group** | *C#, AutoCAD, Revit* Jan 2023 - Apr 2023

- Conducted quality control inspections in Revit for electrical floor plan layouts for Sick Kids Hospital, resulting in a 70% reduction in completion time per layout.
- Increased CAD workflow efficiency by 10% through the development of macros and tools in C#.
- Designed and repaired plumbing and electrical fixtures in Revit for TD Bank floor plan layouts.

## Projects

**FPGA Audio Player** | *C, Quartus Prime, Altera NIOS II* Jul 2026

- Designed an embedded audio player on an Altera NIOS II FPGA, integrating MicroSD, LCD, pushbuttons, hardware timers, and FPGA audio peripherals.
- Implemented FatFs/FAT32 file-system functionality to enumerate WAV files and stream 16-bit stereo audio from MicroSD to left/right audio FIFOs with appropriate buffering.
- Developed interrupt-driven pushbutton controls with hardware-assisted debouncing, implementing user interface functionality without software delays.

**4 Bit ALU PCB** | *Altium Designer, LTspice* Aug 2026 - Present

- Designing a 4-bit CMOS ALU at the transistor level using 28 transistor full-adder cells.
- Implementing binary arithmetic, logic and bit-shift operations with configurable operation selection.
- Prototyped and validated a one-bit full adder on breadboards before scaling the design to four bits.

**Breathalyzer Device** | *C++, STM32, I2C* Sep 2023 - Nov 2023

- Architected a breathalyzer device to approximate the alcohol concentration in the air using an MQ3 alcohol sensor and an STM32 Nucleo-F401RE microcontroller, displaying the result on an LCD screen.
- Implemented a calibration algorithm using data from changes in electron flow within the sensor for more precise readings in various environments.

**Multivibrator PCB** | *Altium Designer* Feb 2025

- Developed an astable multivibrator PCB in Altium Designer for stable pulse generation, including schematic design, component selection, PCB layout optimization and analog waveform analysis.

**Chess Desktop App** | *Python* Jan 2024 - Mar 2024

- Developed a chess game in Python that adheres to traditional rules using the PyGame library.
- Incorporates a simple AI engine built using a minimax algorithm with alpha-beta pruning to play against.
- Employs a recursive algorithm to announce moves played by the player and engine synchronously.

## Education

**University of Waterloo - *BASc, Honours Electrical Engineering*** Apr 2029

Relevant Courses: Embedded Microprocessor Systems, Electronic Circuits, Systems Programming & Concurrency